

Beyond Like-for-Like: Linking Player Similarity with Post-Transfer Playing-Time Retention

Soccer

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Abstract.

Football recruitment often begins with two related but distinct judgments: finding players whose on-ball and off-ball behaviour resembles a target profile, and estimating how much playing time a candidate may retain after a transfer. We evaluate these tasks separately because stylistic resemblance and post-move retention describe different uncertainties, and combining them into one score can hide where each signal becomes unreliable.

The similarity module represents each player-club-competition-season through 32 per-90 behavioural features covering passing, progression, creation, carrying, shooting, defending and recoveries. Features are robustly scaled within competition, and position-matched candidates are ranked using equal-feature Manhattan distance. With no expert-labelled equivalence set, evaluation uses next-season self-retrieval as a temporal consistency proxy rather than a claim of player interchangeability.

Stricter playing-time requirements sharpen retrieval but shrink the pool

Same-player next-season retrieval at three minimum-minutes thresholds. The 900-minute rule, highlighted, is the study's operating point.

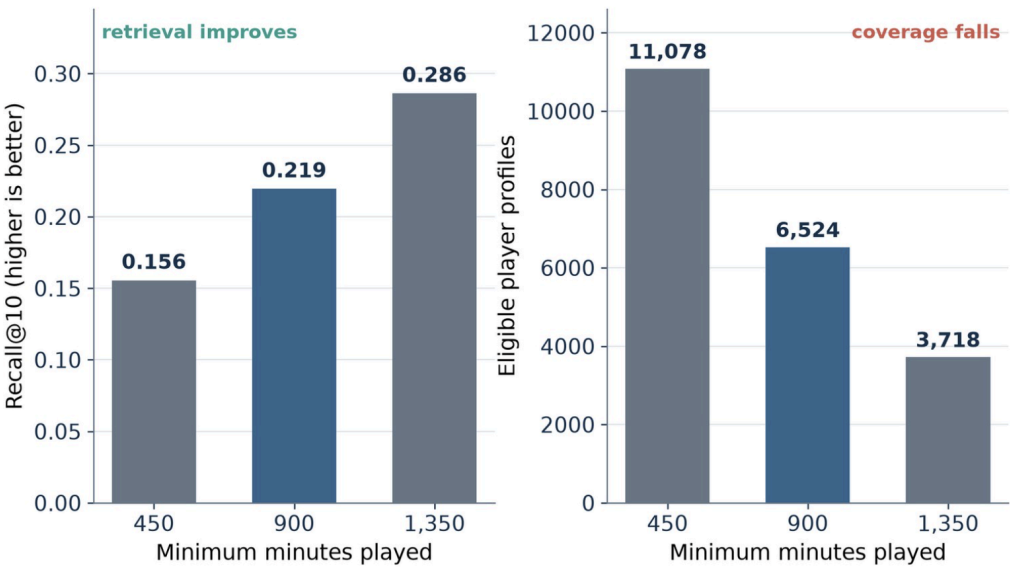


Figure 1. More playing-time evidence improves similarity retrieval while reducing the number of eligible player profiles.

At the 900-minute operating point, 6,524 held-out queries achieve Recall@10 of 21.9% (95% CI 0.209-0.229), compared with roughly 0.2% chance in a median pool of 5,884 candidates. More playing-time evidence improves retrieval but reduces coverage. Context matters sharply: Recall@10 falls from 0.305 when club and competition are unchanged to 0.139 after a club change and 0.066 when both change. Goalkeepers reach 0.099 versus 0.237 for outfield players.

The retention module treats each transfer as a dated event, using 365-day windows before and after the move and at least 450 minutes in each. It predicts the log ratio of post- to pre-transfer minutes from pre-transfer information only. Ordinary least squares is fitted on chronologically earlier, player-disjoint transfers and compared with four pre-set baselines. Across 1,724 held-out transfers, the full model records MAE 0.433 (95% CI 0.417-0.448) and Spearman correlation 0.638, reducing error by 9.66% (95% CI 7.8-11.6%) against the strongest pre-registered baseline at 0.479. A post-hoc reference using prior playing time alone reaches 0.452, leaving a smaller 4.3% gain for richer context. Error rises 6.9% from validation, mean bias is +0.141, and 19 of the 20 largest misses are sharp declines that the model overpredicts.

Most of the error reduction comes from pre-transfer playing time alone

1,724 held-out transfers. Grey and blue bars are the four baselines fixed before the evaluation; the amber bar was added afterwards as a reference.

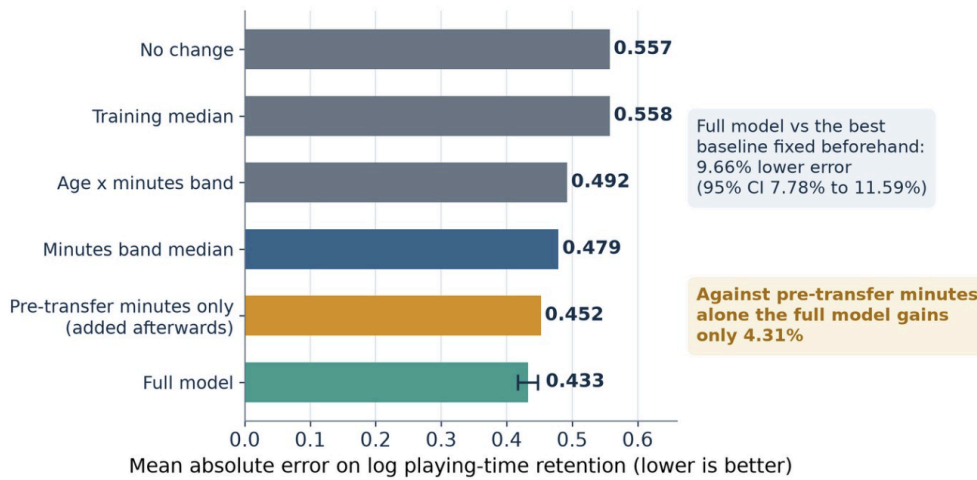


Figure 2. Held-out transfer error for the full model, pre-set baselines and a post-hoc playing-time reference.

Together, the modules support a deliberately separated recruitment workflow: behavioural similarity can organize a stylistic shortlist, while the retention model adds a distinct signal about likely playing-time continuity after a move. Scouts and decision-makers can then combine those outputs with tactical, medical, contractual and financial evidence. The results do not establish causal destination effects, automatic signing decisions or player equivalence, and similarity retrieval is least stable under the context changes most relevant to recruitment.

This research was developed in collaboration with SoccerSolver. SoccerSolver currently works with more than 10 football clubs.

Open-source repository: <https://github.com/RupayanHalder39/MITSloanProblem3-Player-Similarity-Transfer-Retention>